

a comprehensive real-time temperature control. There are models out there from the rudimentary economical ones that rely on thermostat based analogue temperature control to the most sophisticated industrial models that have precision temperature controls and flaunt great looks.

Hand piece. Depending on what you already own and what your soldering needs are, you may require a hand piece. Certain soldering systems come with enough flexibility to allow you to interchange and use a variety of hand pieces to help achieve close soldering blends. A pencil hand piece is usually excellent for SMT and PTH work.

De-soldering iron. For de-soldering, you would need a de-soldering iron, but not all systems come equipped with a de-soldering iron. If you need one, look for a solder station that includes a well-designed de-soldering iron and a versatile hand-piece stand.

Soldering and de-soldering bits. To match really small or heavy-duty applications, and to meet your soldering and de-soldering requirements and needs, a variety of soldering and de-soldering bits are needed. So always go for a rework station that includes a versatile set of bits to match your work needs, and make sure the quality and re-availability of these bits are taken care of.

Special attachments. For the sake of convenience during work, and versatility, choose a model that comes with special attachments, say, attachments for the removal of QFP packages, a vacuum picker and so on.

Vacuum pump. Always go for a vacuum pump having good suction level. Read manufacturer's specifications carefully at the time of purchase. A dual-reflow-type vacuum pump with a release valve is a good choice. A comprehensive vacuum-level indicator can further make your life easy during SMT and PTH work.

Thermal tweezers. Recent advanced rework stations are available with special tools like thermal tweezers and thermal peelers. Consequently, review your needs before buying your rework unit.

Safety features. As with any soldering equipment purchase, ensure that the equipment includes a variety of safety features. It should be ESD-safe, have an accurate temperature control and a temperature lock function besides a tip-cleaning patch to ensure proper solder cleanup.



Fig. 3: ESD tools



Fig. 4: Using ESD-safe wrist and ankle straps

Versatility. When choosing a trusted brand for your soldering station, you have the added benefit of versatility. Many of the soldering equipment come with interchangeable instrument attachments, allowing for the biggest range of soldering applications.

Wattage. It holds prime importance in making your buying decision for a rework station. Unless you are going for repair of instruments involving high-profile, rugged components, a 140W or equivalent is quite reasonable and goes well for almost all regular work.

Soldering torch. An acetylene or

butane based torch attachment is particularly an incentive if you are interested in high-profile assembly work that involves metal soldering of assorted types, chassis work or work involving galvanised wires.

ESD tools and devices. Modern-day work involves electrostatic-sensitive components on a large scale. To work with such devices, normal equipment is quite unsafe. Need for special ESD-safe tools and devices is mandatory. We recommend use of the following devices if you are involved in handling and use of such devices:

- ESD-safe mats
- ESD-safe tool kits
- Anti-static dispensers
- ESD-safe wrist and ankle straps
- Periodic verification tester
- Precision surface resistance monitor
- ESD field-strength meter
- Automatic ground-fault watch dog
- Temporal ground extenders

Important note

EFY team would like to suggest some mandatory safety devices as well.

A proper electrical ground is the foremost requirement in any assembly or repair workshop, but unfortunately, it is often neglected. Soldering fumes are very toxic in nature, so you must install a fume exhauster in your workshop.

In workshops doing high-profile work involving torch soldering, special fire retardant gowns and goggles should be put to use to avoid any untoward incident.

Proper layout of power cables can be very helpful to avoid accidents to a large extent. Fire or arc light monitors and alarms should be employed in all assembly lines involving torch soldering.

A first aid burn kit comes in handy for minor injuries during soldering jobs. So keep one in your tool kit. **EFY**